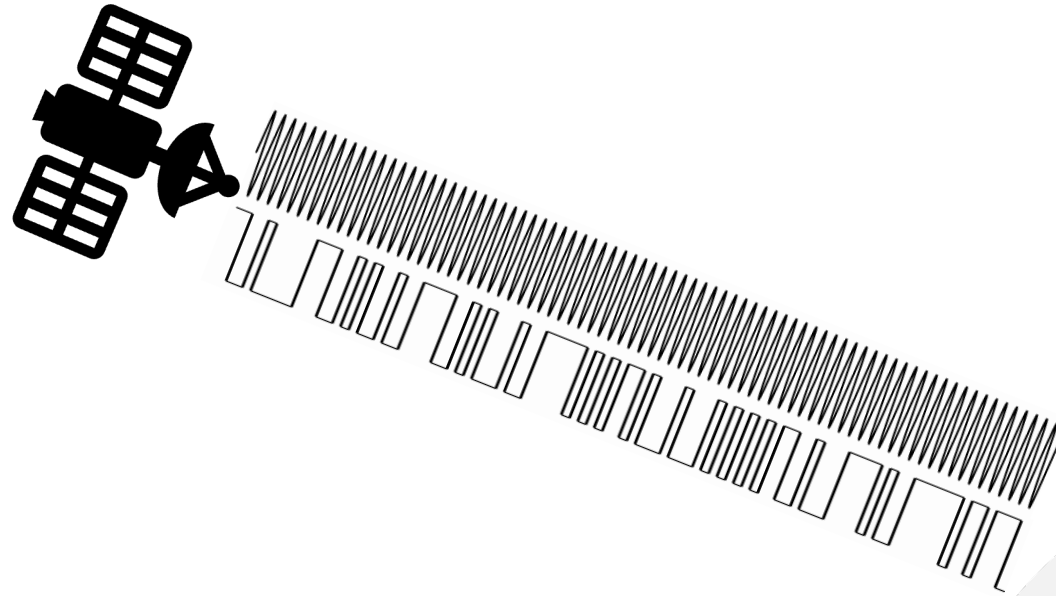


ASEN 5090 Lecture 7

GPS Measurements & IGS Data



Penny Axelrad

Penina.Axelrad@Colorado.edu

(303) 492-6872



Next Section of the Course

We are now moving on to Part II of the textbook

– Estimation of Position, Velocity, and Time

- **Ch 5 – Measurements & Error Sources**
- Ch 6 - PVT Estimation (including appendix w/generic LS solution)
- Ch 7 - Precise positioning



Outline

- GPS pseudorange or code measurements
- GPS carrier phase measurements
- Code and carrier measurement models

Reading Assignment

- Ch 5.1-5.2

Questions

- How does GPS measure time of arrival?
- What are the different types of measurements logged in RINEX observation files?

Measurement & Prediction Errors

- GEOMETRIC RANGE (GPS ORBIT ERRORS)
- IONOSPHERE (FREE ELECTRONS)
- TROPOSPHERE (NEUTRAL ATMOSPHERE)
- MULTIPATH
- RELATIVITY (NOT SHOWN IN EQUATIONS)
- CLOCKS
- MEASUREMENT NOISE



OBSERVABLES / MEASUREMENT TYPES

- L1 pseudorange – C/A or P(Y)
- L2 pseudorange - P(Y) or C2
- L5 pseudorange – L5C
- Carrier Phase & Doppler
- Signal to noise ratio (SNR)



BASIC GPS MEASUREMENT: PSEUDORANGE

- Receiver measures difference between time of transmission and time of reception based on correlation of received signal with a local replica

$$\rho = c(t_u - t^s)$$

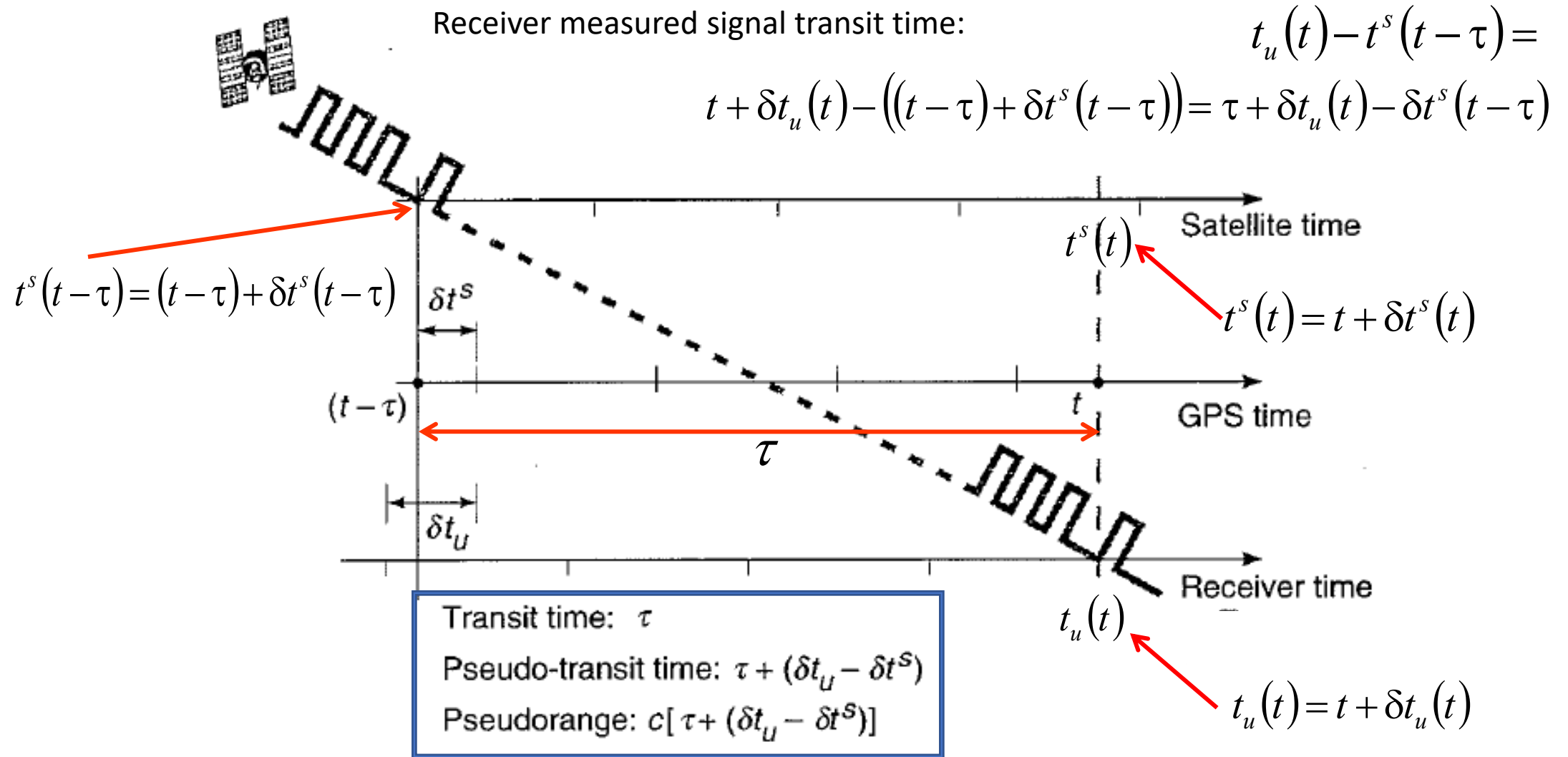
t_u = time of reception as observed by the receiver

t^s = time of transmission as generated by the satellite

The measured pseudorange is not the true range between the satellite and receiver. That is what we clarify with the measurement model/observable equation.



Code Phase & Pseudorange



Pseudorange measurements

$$\rho(t) = c\tau + c(\delta t_u(t) - \delta t^s(t - \tau)) + \boxed{\varepsilon_\rho(t)}$$

- unmodeled effects
- modeling errors
- measurement errors

$$c\tau = r(t, t - \tau) + I_\rho(t) + T_\rho(t)$$

Geometric range between
the user position at t
and SV position at $t - \tau$

Ionosphere
delay

Troposphere delay

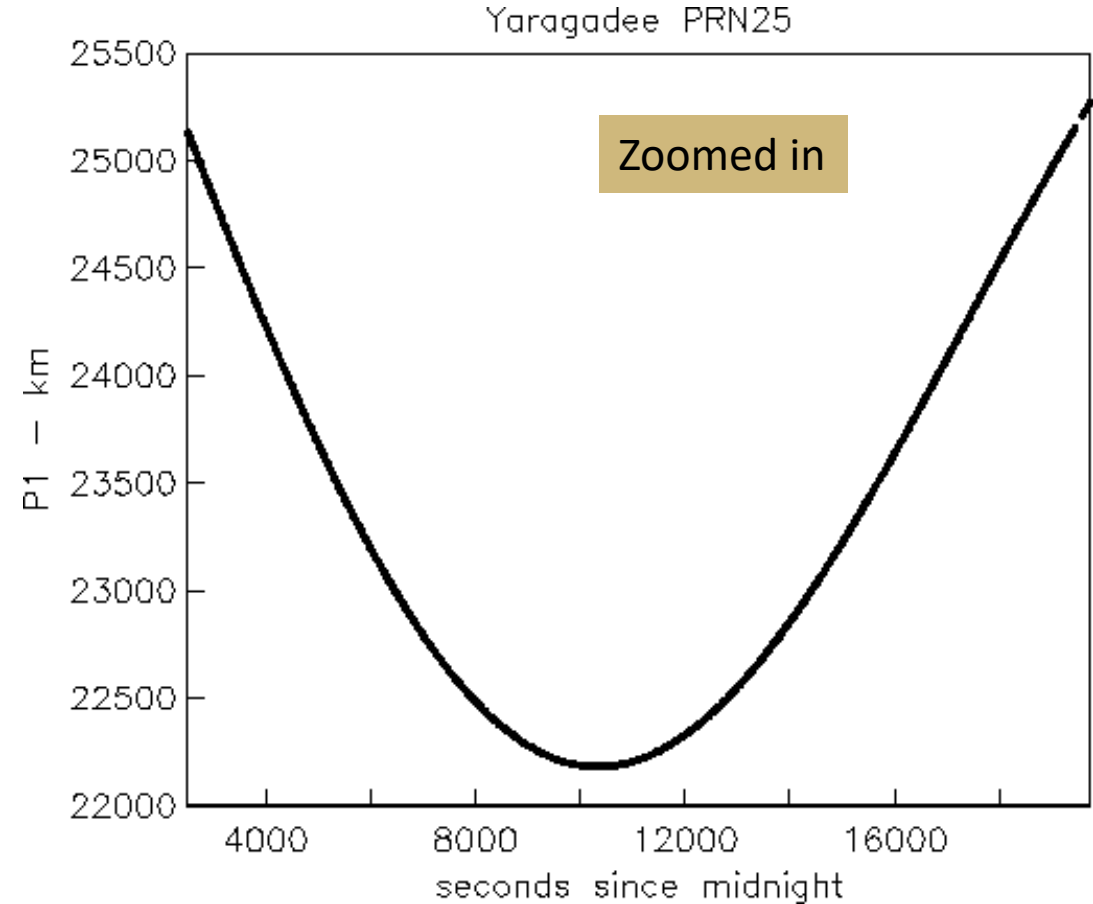
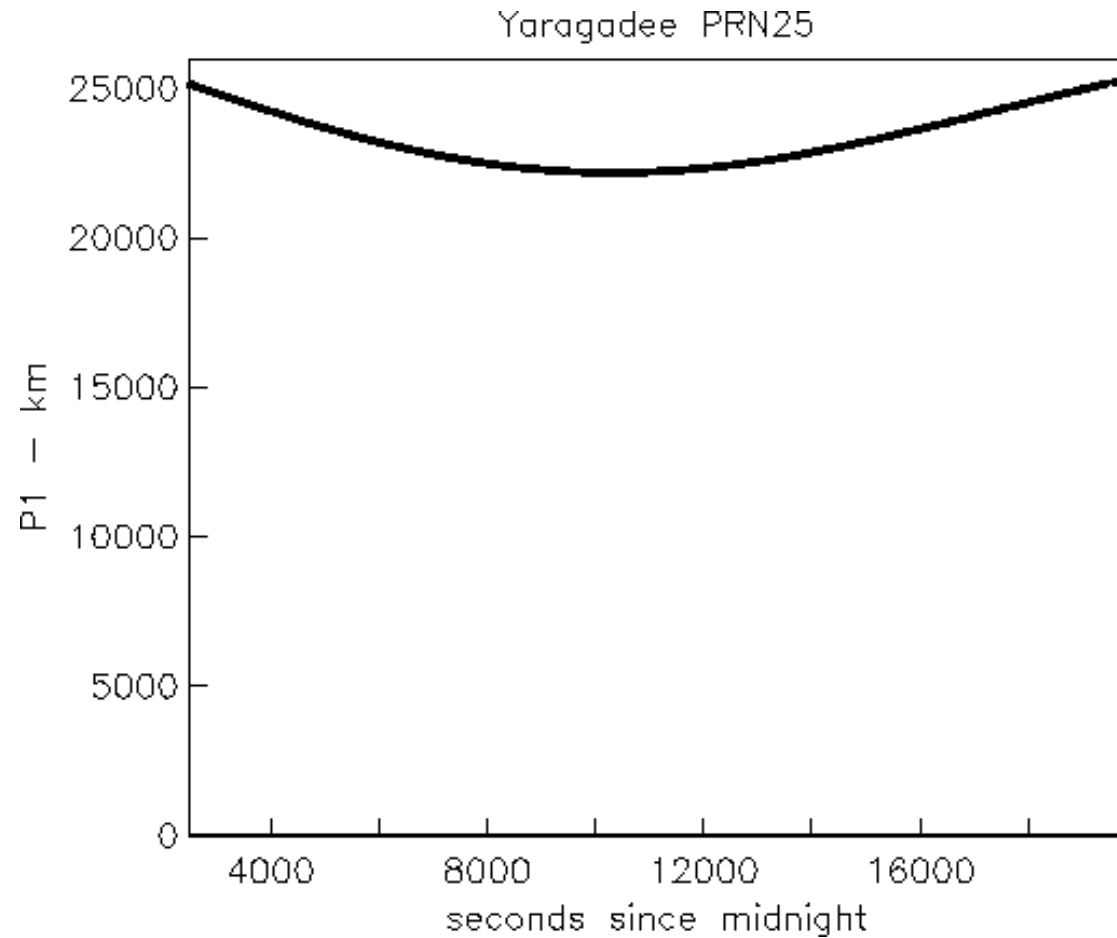
$$\rho(t) = r(t, t - \tau) + I_\rho(t) + T_\rho(t) + c(\delta t_u(t) - \delta t^s(t - \tau)) + \varepsilon_\rho(t)$$

Accepted conventional notation:

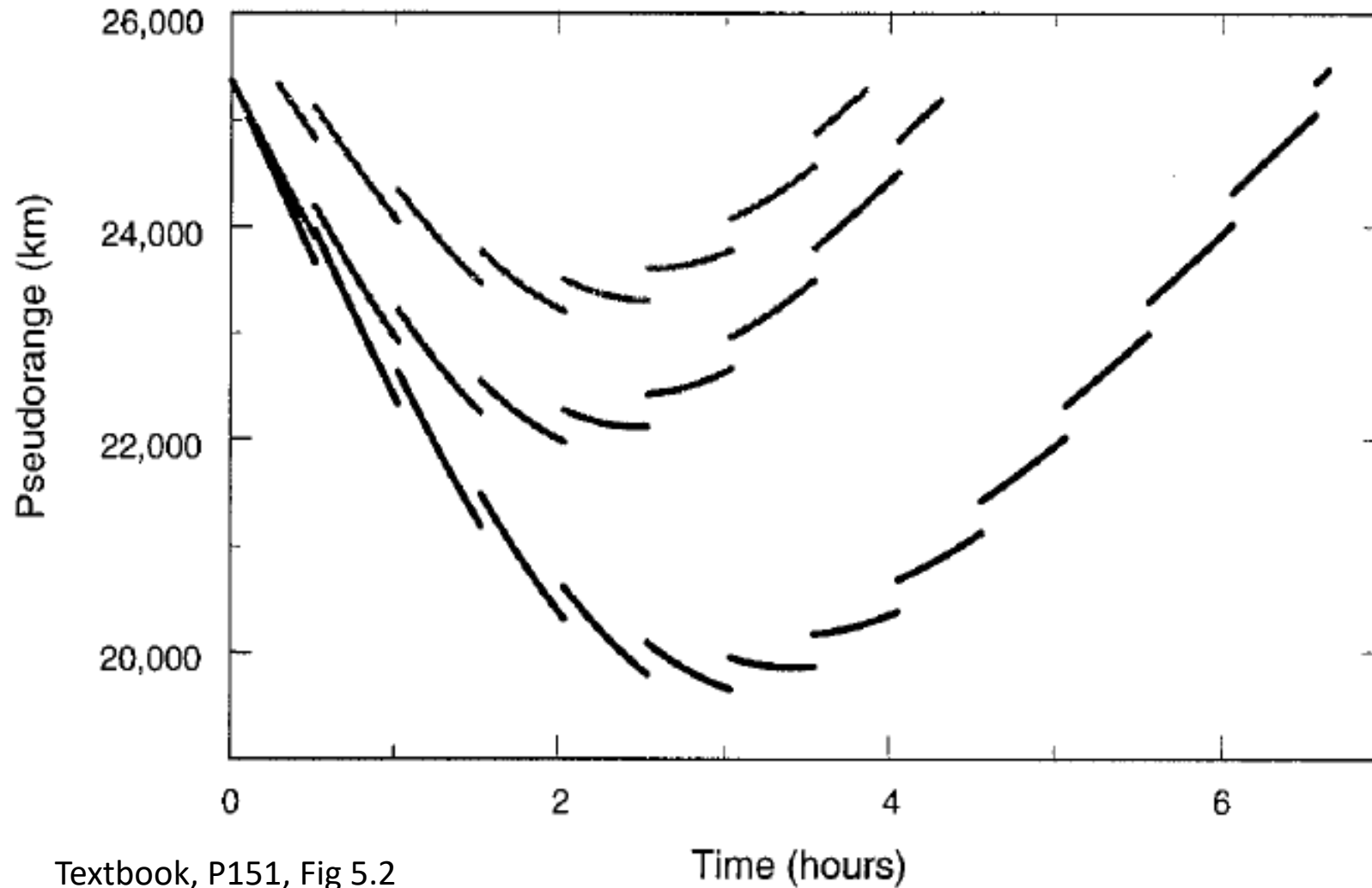
$$\rho = r + I_\rho + T_\rho + c(\delta t_u - \delta t^s) + \varepsilon_\rho$$



Example of Pseudorange Measurements



What kind of error do we see here?

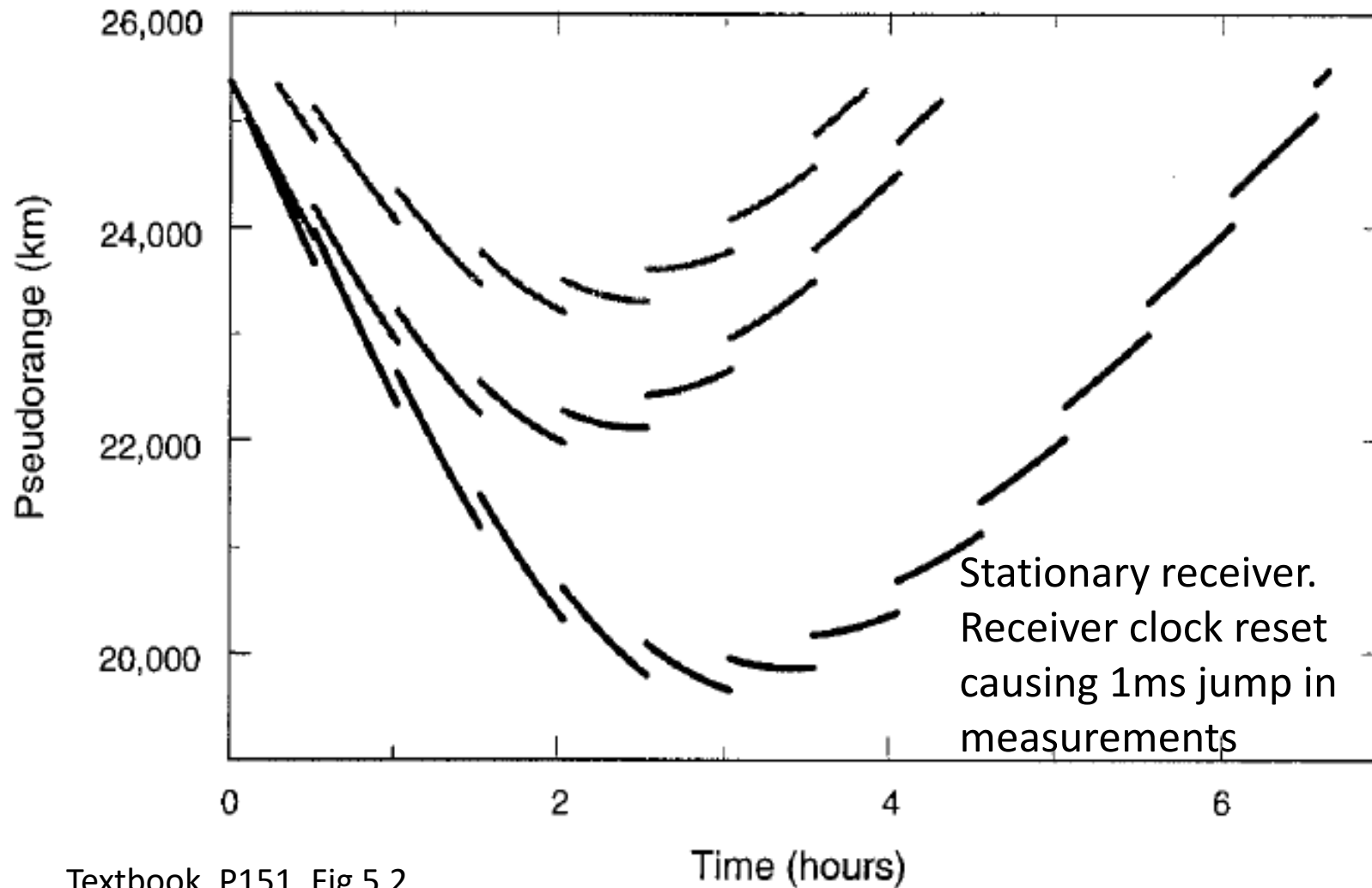


Pseudorange measurements for 3 different satellites from a stationary ground receiver

Textbook, P151, Fig 5.2



Receiver Clock Drift



Questions:

Is the clock running fast or slow?

Nominal frequency = 10.23MHz

What is the frequency offset?

Textbook, P151, Fig 5.2

Question Answers

$$\frac{\Delta t}{T} = -\frac{\Delta f_{\text{error}}}{f_0} \quad \sim 5.6\text{e-}7$$

Based on the figure, a 1ms jump occurs at about every 30 minutes (the clock has to be moved forward 1ms every 30 min, it is too slow):

$$\Delta f = -f_0 \frac{\Delta t}{T} = -10.23(\text{MHz}) \times \frac{10^{-3}}{30 \times 60} = -5.68(\text{Hz})$$

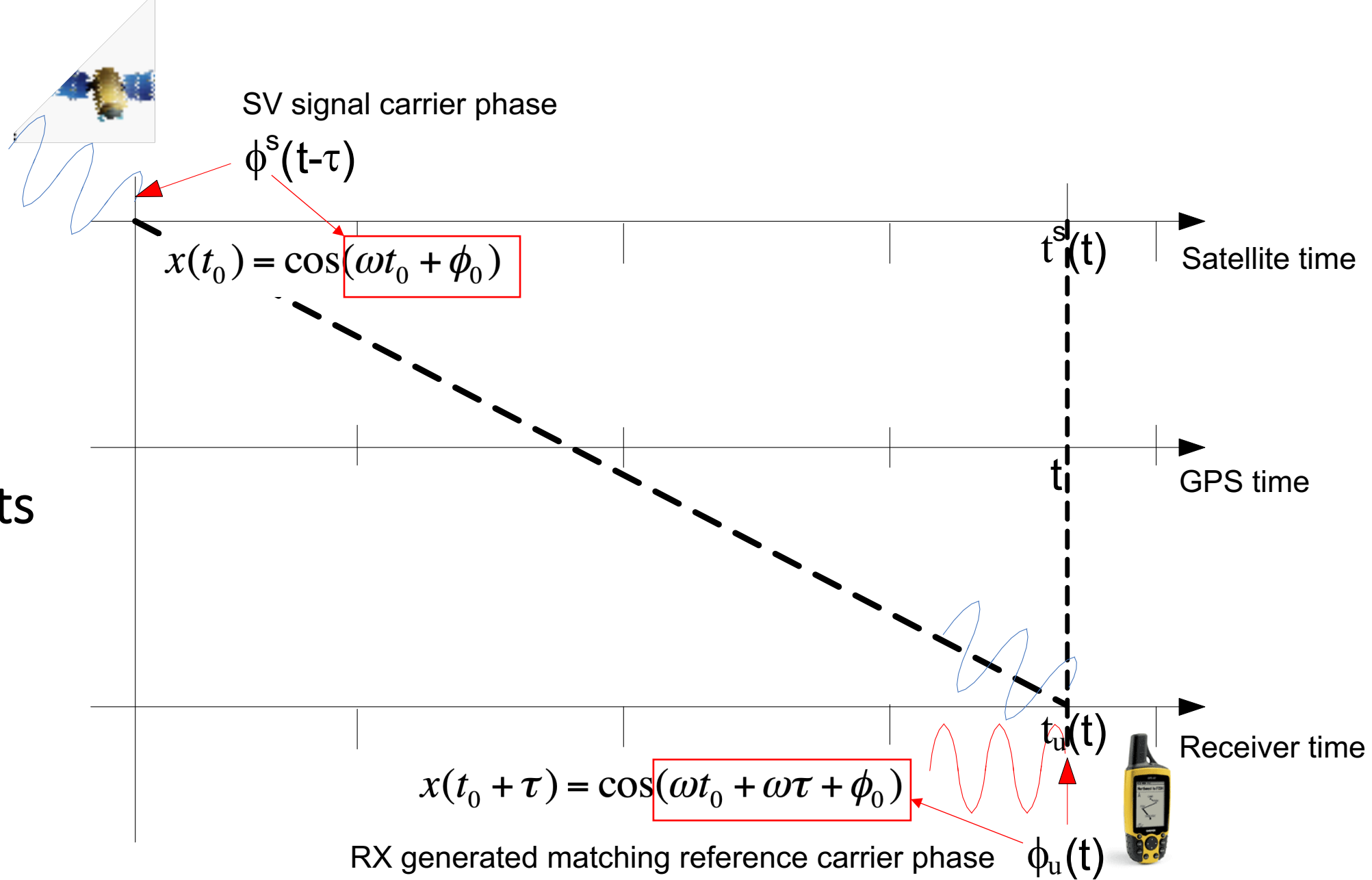
The oscillator frequency is 5.7 Hz smaller than its nominal value.



Carrier Phase Measurements

$$\phi(t) = \frac{c\tau}{\lambda} + N :$$

in unit of cycles



CARRIER PHASE MEASUREMENT

- Receiver accumulates changes in carrier frequency of received signal
- This accumulation provides a precise measure of the change in range to the satellite (ADR or accumulated delta range).
- This means that carrier phase data have an unknown bias that must be estimated for the data to be used.

Carrier Phase Measurement Under Realistic Conditions

$$\phi(t) = \frac{c\tau}{\lambda} + N = \frac{1}{\lambda} \left(r(t, t - \tau) + I_\phi + T_\phi + c\delta t_u - c\delta t^s \right) + \varepsilon_\phi + N$$

Integer cycle ambiguity

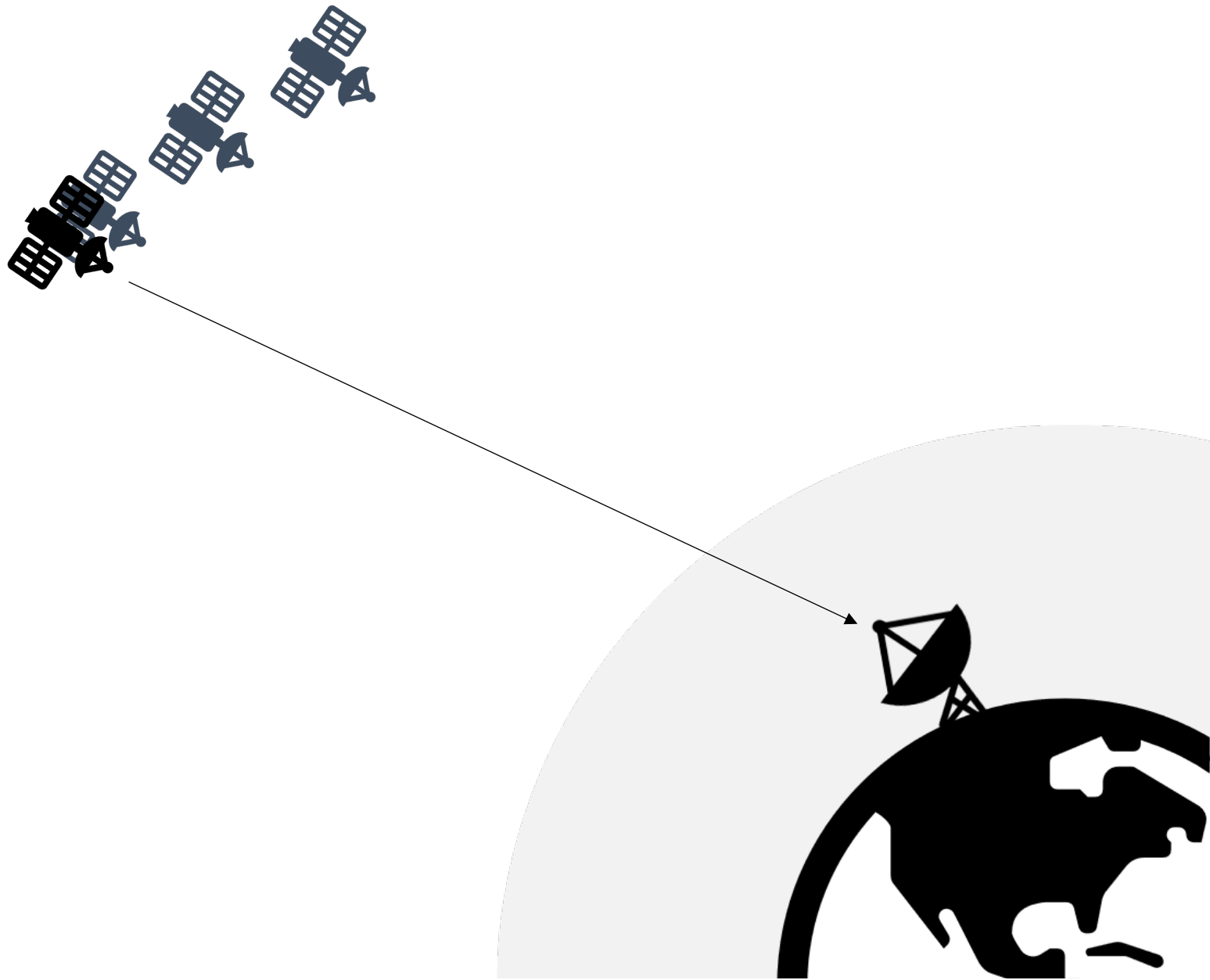
Integrated Doppler (delta pseudorange):

Carrier phase differences over a short time interval



GEOMETRIC RANGE

- 1) Compute the GPS satellite position in ECEF at T_r based on the ***broadcast ephemeris*** or final orbits (sp3).
- 2) Use an *a priori* value for the receiver coordinates (RX) to find geometric range $R = |r_{GPS} - r_{RX}|$.
- 3) Compute the time of transmission:
 $T_t = T_r - R/c$ (use true speed of light, NOT $3e8$)
- 4) Compute the satellite position at T_t in ECEF at T_t based on the broadcast ephemeris.
- 5) Rotate the satellite position to ECEF at T_r .
- 6) Compute a new geometric range using this position for r_{GPS} , $R = |r_{GPS} - r_{RX}|$.
- 7) Repeat steps 3-6 until convergence.
(If it takes more than two iterations, something is wrong).



Broadcast Ephemeris

- Each satellite transmits only its own “broadcast ephemeris” data.
- A receiver decodes this data & uses it to compute the location of the GPS satellite. This is required for positioning, navigation and timing.

```
NIST00USA_R_20252230000_01D_GN.rnx
3.04      NAVIGATION DATA      G      RINEX VERSION / TYPE
sbfin-13.8.0 20250812 003027 UTC PGM / RUN BY / DATE
gfzrnx-1.13.7761 FILE PROCESSING 20250812 003059 UTC COMMENT
GPSA 1.3039E-08 2.2352E-08 -5.9605E-08 -1.1921E-07 IONOSPHERIC CORR
GPSB 1.0650E+05 1.3107E+05 -6.5536E+04 -2.6214E+05 IONOSPHERIC CORR
GPUT -1.8626451492E-09-5.329070518E-15 405504 2379 TIME SYSTEM CORR
18 LEAP SECONDS
END OF HEADER
G01 2025 08 11 00 00 00 3.315396606922e-04 6.025402399246e-12 0.000000000000e+00
7.100000000000e+01 7.481250000000e+01 4.095170580292e-09-3.011531052779e+00
3.954395651817e-06 6.326901493594e-04 1.169741153717e-05 5.153717447281e+03
8.640000000000e+04 3.539025783539e-08 4.901822356577e-01 1.862645149231e-09
9.594239576417e-01 1.540000000000e+02 1.058330939046e-01-7.877828143163e-09
1.071473202588e-10 1.000000000000e+00 2.379000000000e+03 0.000000000000e+00
2.000000000000e+00 0.000000000000e+00-8.847564458847e-09 7.100000000000e+01
7.921800000000e+04 4.000000000000e+00 0.000000000000e+00 0.000000000000e+00
G01 2025 08 11 02 00 00 3.315825015306e-04 6.025402399246e-12 0.000000000000e+00
7.200000000000e+01 6.590625000000e+01 4.100170788571e-09-1.961933775628e+00
3.580003976822e-06 6.340456893668e-04 1.215003430843e-05 5.153715024948e+03
9.360000000000e+04-7.636845111847e-08 4.901263360949e-01 7.823109626770e-08
9.594247929680e-01 1.453750000000e+02 1.063880884509e-01-7.787824394146e-09
2.607251459631e-11 1.000000000000e+00 2.379000000000e+03 0.000000000000e+00
2.000000000000e+00 0.000000000000e+00-8.847564458847e-09 7.200000000000e+01
8.641800000000e+04 4.000000000000e+00 0.000000000000e+00 0.000000000000e+00
```

IS-GPS-200

Table 20-II. Ephemeris Data Definitions

M_0	Mean Anomaly at Reference Time
Δn	Mean Motion Difference From Computed Value
e	Eccentricity
$\sqrt{A}$	Square Root of the Semi-Major Axis
Ω_0	Longitude of Ascending Node of Orbit Plane at Weekly Epoch
i_0	Inclination Angle at Reference Time
ω	Argument of Perigee
$\dot{\Omega}$	Rate of Right Ascension
IDOT	Rate of Inclination Angle
C_{uc}	Amplitude of the Cosine Harmonic Correction Term to the Argument of Latitude
C_{us}	Amplitude of the Sine Harmonic Correction Term to the Argument of Latitude
C_{rc}	Amplitude of the Cosine Harmonic Correction Term to the Orbit Radius
C_{rs}	Amplitude of the Sine Harmonic Correction Term to the Orbit Radius
C_{ic}	Amplitude of the Cosine Harmonic Correction Term to the Angle of Inclination
C_{is}	Amplitude of the Sine Harmonic Correction Term to the Angle of Inclination
t_{oe}	Reference Time Ephemeris (reference paragraph 20.3.4.5)
IODF	Issue of Data (Ephemeris)

Navigation data from RINEX nav file

Values scaled and units converted from original transmission

A6 GNSS Navigation Message File – GPS Data Record Description

TABLE A6 GNSS NAVIGATION MESSAGE FILE – GPS DATA RECORD DESCRIPTION		
NAV. RECORD	DESCRIPTION	FORMAT
<i>SV / EPOCH / SV CLK</i>	- Satellite system (G), sat number (PRN) - Epoch: Toc - Time of Clock (GPS) year (4 digits) - month, day, hour, minute, second - SV clock bias (seconds) - SV clock drift (sec/sec) - SV clock drift rate (sec/sec2)	A1,I2.2, 1X,I4, 5(1X,I2.2), 3D19.12 (*)
BROADCAST ORBIT - 1	- IODE Issue of Data, Ephemeris - Crs (meters) - Delta n (radians/sec) - M0 (radians)	4X,4D19.12 (***)
BROADCAST ORBIT - 2	- Cuc (radians) - e Eccentricity - Cus (radians) - sqrt(A) (sqrt(m))	4X,4D19.12
BROADCAST ORBIT - 3	- Toe Time of Ephemeris (sec of GPS week) - Cic (radians) - OMEGA0 (radians) - Cis (radians)	4X,4D19.12
BROADCAST ORBIT - 4	- i0 (radians) - Cre (meters) - omega (radians) - OMEGA DOT (radians/sec)	4X,4D19.12
BROADCAST ORBIT - 5	- IDOT (radians/sec) - Codes on L2 channel - GPS Week # (to go with TOE) Continuous number, not mod(1024)! - L2 P data flag	4X,4D19.12
BROADCAST ORBIT - 6	- SV accuracy (meters) See GPS ICD Section 20.3.3.3.1.3 use specified equations to define nominal values, N = 0-6: use $2^{(1+N/2)}$ (round to one decimal place i.e. 2.8, 5.7 and 11.3), N= 7-15:use $2^{(N-2)}$, 8192 specifies use at own risk - SV health (bits 17-22 w 3 sf 1) - TGD (seconds) - IODC Issue of Data, Clock	4X,4D19.12
BROADCAST ORBIT - 7	- Transmission time of message (sec of GPS week, derived e.g. from Z-count in Hand Over Word (HOW)) - Fit Interval in hours; bit 17 w 10 sf 2 + IODC & Table 20-XII of GPS ICD. (BNK if Unknown). - Spare(x2) (see Section 6.4)	4X,4D19.12



Table 20-IV. Broadcast Navigation User Equations (sheet 1 of 4)

$\mu = 3.986005 \times 10^{14} \text{ meters}^3/\text{sec}^2$	WGS 84 value of the earth's gravitational constant for GPS user
$\dot{\Omega}_e = 7.2921151467 \times 10^{-5} \text{ rad/sec}$	WGS 84 value of the earth's rotation rate
$A = \left(\sqrt{A} \right)^2$	Semi-major axis
$n_0 = \sqrt{\frac{\mu}{A^3}}$	Computed mean motion (rad/sec)
$t_k = t - t_{oe}^*$	Time from ephemeris reference epoch
$n = n_0 + \Delta n$	Corrected mean motion
$M_k = M_0 + nt_k$	Mean anomaly
	Kepler's equation ($M_k = E_k - e \sin E_k$) may be solved for Eccentric anomaly (E_k) by iteration:
$E_0 = M_k$	– Initial Value (radians)
$E_j = E_{j-1} + \frac{M_k - E_{j-1} + e \sin E_{j-1}}{1 - e \cos E_{j-1}}$	– Refined Value, minimum of three iterations, (j=1,2,3)
$E_k = E_j$	– Final Value (radians)
$v_k = 2 \tan^{-1} \left(\sqrt{\frac{1+e}{1-e}} \tan \frac{E_k}{2} \right)$	True Anomaly (unambiguous quadrant)
* t is GPS system time at time of transmission, i.e., GPS time corrected for transit time (range/speed of light). Furthermore, t_k shall be the actual total time difference between the time t and the epoch time t_{oe}, and must account for beginning or end of week crossovers. That is, if t_k is greater than 302,400 seconds, subtract 604,800 seconds from t_k. If t_k is less than -302,400 seconds, add 604,800 seconds to t_k.	

Table 20-IV. Broadcast Navigation User Equations (sheet 2 of 4)

$\Phi_k = v_k + \omega$		Argument of Latitude
$\delta u_k = c_{us}\sin 2\Phi_k + c_{uc}\cos 2\Phi_k$	Argument of Latitude Correction	} Second Harmonic
$\delta r_k = c_{rs}\sin 2\Phi_k + c_{rc}\cos 2\Phi_k$	Radius Correction	
$\delta i_k = c_{is}\sin 2\Phi_k + c_{ic}\cos 2\Phi_k$	Inclination Correction	
$u_k = \Phi_k + \delta u_k$		Corrected Argument of Latitude
$r_k = A(1 - e \cos E_k) + \delta r_k$		Corrected Radius
$i_k = i_0 + \delta i_k + (\text{IDOT}) t_k$		Corrected Inclination
$\left. \begin{array}{l} x'_k = r_k \cos u_k \\ y'_k = r_k \sin u_k \end{array} \right\}$		Positions in orbital
$\Omega_k = \Omega_0 + (\dot{\Omega} - \dot{\Omega}_e) t_k - \dot{\Omega}_e t_{oe}$		Corrected longitude of ascending node.
$\left. \begin{array}{l} x_k = x'_k \cos \Omega_k - y'_k \cos i_k \sin \Omega_k \\ y_k = x'_k \sin \Omega_k + y'_k \cos i_k \cos \Omega_k \\ z_k = y'_k \sin i_k \end{array} \right\}$		Earth-fixed

Table 20- IV. Broadcast Navigation User Equations (sheet 3 of 4)

Element/Equation	Description
SV Velocity	
$\dot{E}_k = n / (1 - e \cos E_k)$	Eccentric Anomaly Rate
$\dot{\nu}_k = \dot{E}_k \sqrt{1 - e^2} / (1 - e \cos E_k)$	True Anomaly Rate
$(di_k / dt) = (IDOT) + 2 \dot{\nu}_k (c_{is} \cos 2\phi_k - c_{ic} \sin 2\phi_k)$	Corrected Inclination Angle Rate
$\dot{u}_k = \dot{\nu}_k + 2\dot{\nu}_k (c_{us} \cos 2\phi_k - c_{uc} \sin 2\phi_k)$	Corrected Argument of Latitude Rate
$\dot{r}_k = eA\dot{E}_k \sin E_k + 2\dot{\nu}_k (c_{rs} \cos 2\phi_k - c_{rc} \sin 2\phi_k)$	Corrected Radius Rate
$\dot{\Omega}_k = \dot{\Omega} - \dot{\Omega}_e$	Longitude of Ascending Node Rate
$\dot{x}'_k = \dot{r}_k \cos u_k - r_k \dot{u}_k \sin u_k$	In- plane x velocity
$\dot{y}'_k = \dot{r}_k \sin u_k + r_k \dot{u}_k \cos u_k$	In- plane y velocity
$\dot{x}_k = -x'_k \dot{\Omega}_k \sin \Omega_k + \dot{x}'_k \cos \Omega_k - \dot{y}'_k \sin \Omega_k \cos i_k - y'_k (\dot{\Omega}_k \cos \Omega_k \cos i_k - (di_k / dt) \sin \Omega_k \sin i_k)$	Earth- Fixed x velocity (m/s)
$\dot{y}_k = x'_k \dot{\Omega}_k \cos \Omega_k + \dot{x}'_k \sin \Omega_k + \dot{y}'_k \cos \Omega_k \cos i_k - y'_k (\dot{\Omega}_k \sin \Omega_k \cos i_k + (di_k / dt) \cos \Omega_k \sin i_k)$	Earth- Fixed y velocity (m/s)
$\dot{z}_k = \dot{y}'_k \sin i_k + y'_k (di_k / dt) \cos i_k$	Earth- Fixed z velocity (m/s)

Table 20-IV. Broadcast Navigation User Equations (sheet 4 of 4)

Element/Equation	Description
SV Acceleration	
$R_E = 6378137.0$ meters	WGS 84 Earth Equatorial Radius
$J_2 = 0.0010826262$	Oblate Earth Gravity Coefficient
$F = - (3/2) J_2 (\mu / r_k^2) (R_E / r_k)^2$	Oblate Earth acceleration Factor
$\ddot{x}_k = -\mu (x_k / r_k^3) + F [(1 - 5 (z_k / r_k)^2)(x_k / r_k)] + 2\dot{y}_k \dot{\Omega}_e + x_k \dot{\Omega}_e^2$	Earth- Fixed x acceleration (m/s ²)
$\ddot{y}_k = -\mu (y_k / r_k^3) + F [(1 - 5 (z_k / r_k)^2)(y_k / r_k)] - 2\dot{x}_k \dot{\Omega}_e + y_k \dot{\Omega}_e^2$	Earth- Fixed y Acceleration (m/s ²)
$\ddot{z}_k = -\mu (z_k / r_k^3) + F [(3 - 5 (z_k / r_k)^2)(z_k / r_k)]$	Earth- Fixed z Acceleration (m/s ²)

20.3.3.4.3.2 Parameter Sensitivity

The sensitivity of the SV's antenna phase center position to small perturbations in most ephemeris parameters is extreme. The sensitivity of position to the parameters $\sqrt{A}$, C_{rc} and C_{rs} is about one meter/meter. The sensitivity of position to the angular parameters is on the order of 10^8 meters/semicircle, and to the angular rate parameters is on the order of 10^{12} meters/semicircle/second. Because of this extreme sensitivity to angular perturbations, the value of π used in the curve fit is given here. π is a mathematical constant, the ratio of a circle's circumference to its diameter.

Here π is taken as

$$\pi = 3.1415926535898.$$

20.3.3.4.3.3 Coordinate Systems

20.3.3.4.3.3.1 ECEF Coordinate System

The equations given in Table 20-IV provide the SV's antenna phase center position in the WGS 84 ECEF coordinate system defined as follows:

IGS and RINEX

ALIC



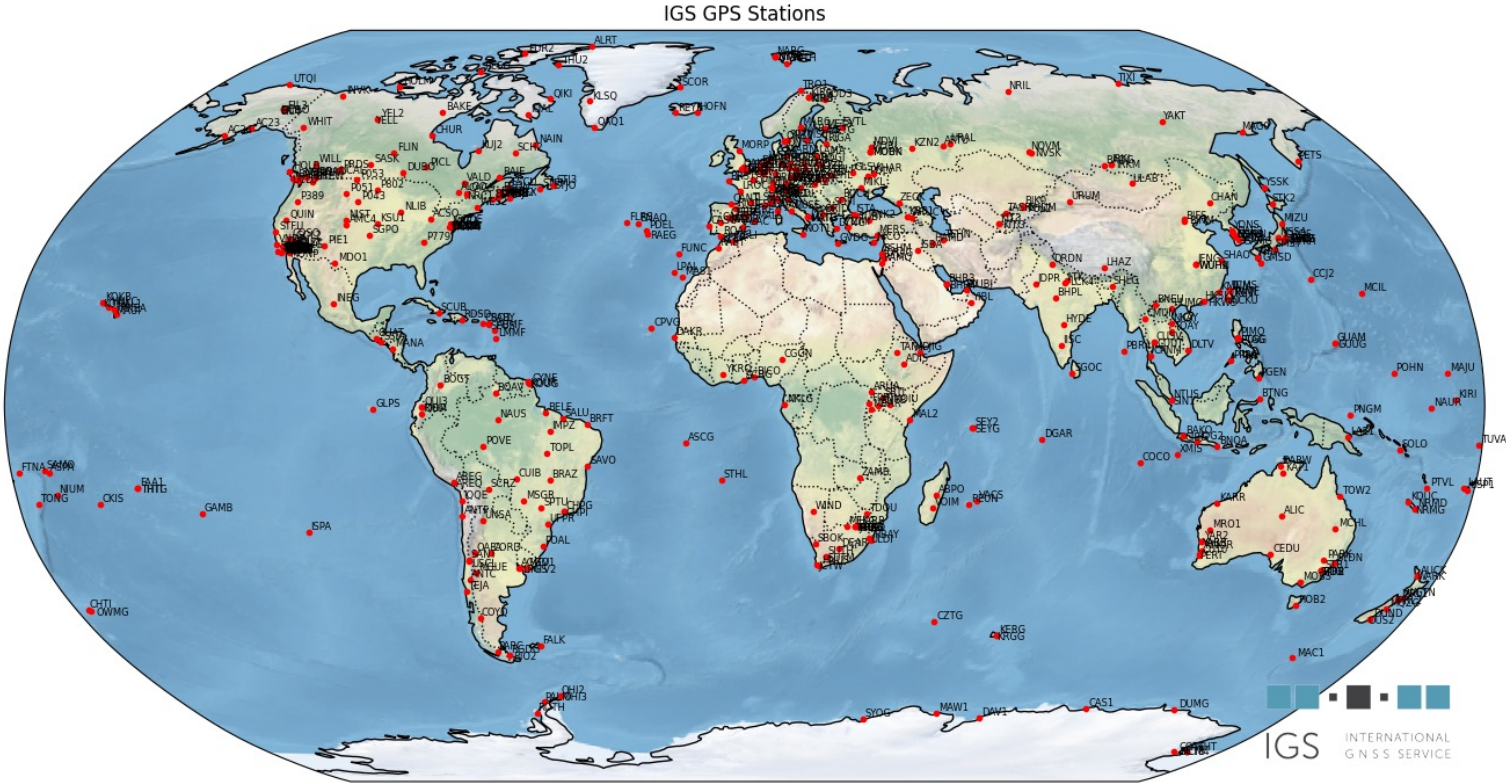
USN7



NIST



RINEX
The Receiver Independent Exchange Format
Version 3.05



IGS - International GNSS Service

Mission: The International GNSS Service provides, on an openly available basis, the highest-quality GNSS data, products and services in support of the terrestrial reference frame, Earth observation and research; positioning, navigation and timing; and other applications that benefit science and society.

- A voluntary federation of agencies, universities, and research institutions in more than 100 countries; working together to provide the highest precision GPS satellite orbits in the world.
- Access to data from > 500 stations
- Analysis Centers develop and produce products to support science
- Supports geodetic research and scholarly publications



<http://www.igs.org/>

imagery ©2019 NASA

IGS – International GNSS Service

<http://www.igs.org>

- A voluntary federation of over 200 self-funding agencies, universities, and research institutions in more than 100 countries; working together to provide the highest precision GPS satellite orbits in the world.
- Providing free and open access to the highest precision products available for scientific advancement and public benefit
- Producing products that support realization of the International Terrestrial Reference Frame while providing access to tracking data from over 400 worldwide reference stations
- Working for the continuous development of new applications and products through Working Groups and Pilot Projects
- Supporting geodetic research and scholarly publications
- Functioning as a component of the Global Geodetic Observing System (GGOS) and member of the World Data System (WDS), with sponsorship from NASA

IGS Site Example

NIST00USA

Latitude, Longitude: 39.995, -105.263

Elevation: 1648.353 m

Country/Region: Boulder, United States of America (the)

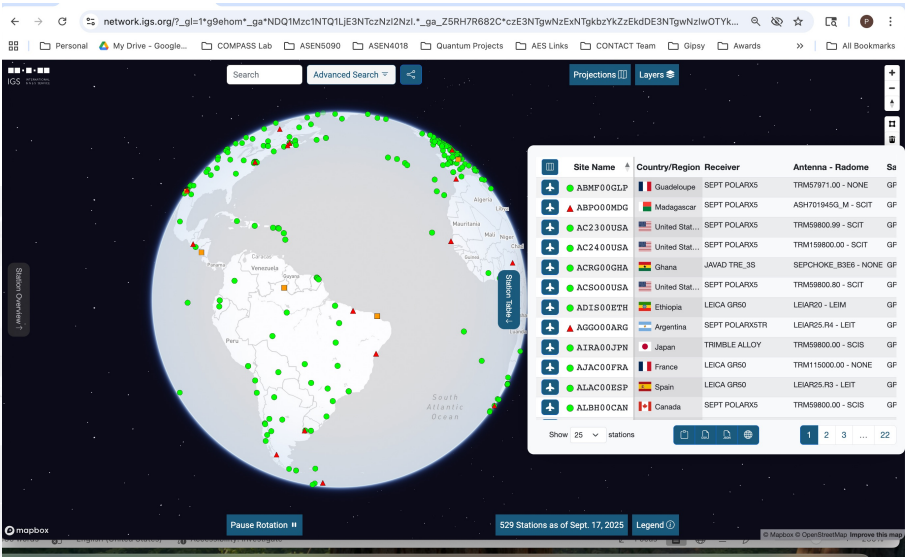
Last Data Available: 2025-09-15

Receiver: SEPT POLARX5TR

Antenna: NOV750.R4

Satellite System: GPS+GLO+GAL+BDS+QZSS

Station Info



NIST00USA

Map Photos

Country/Region: Boulder, United States of America (the)
Latitude, Longitude: 39.995067, -105.262606
Elevation: 1648.353 (m)

Equipment Advisories

Receiver - Firmware	Antenna - Radome	Date
SEPT POLARXSTR - 5.4.0	NOV750.R4 - NONE	June 17, 2022
SEPT POLARXSTR - 5.4.0	NOV702 - NONE	June 15, 2022
NOV OEM4-G2 - 2.200-L1L2W	NOV702 - NONE	May 10, 2006

Station Information Downloads Attachments

Receiver - Firmware: SEPT POLARXSTR - 5.4.0
Antenna - Radome: NOV750.R4 - NONE
Antenna Calibration: ROBOT
Clock: EXTERNAL H-MASER
Last RINEX Data - Primary Data Center: 2025-09-15 (v3) - JPL
Constellation - RINEX: GPS+GLO+GAL+BDS+QZSS+SBAS
Constellation - RealTime:
DOMES Number: 49507M002
Nearby Tide Gauge: N/A

Analysis Center Usage

AC	Final	Rapid	Ultra
EMR	Feb. 16, 2025		June 5, 2024
MIT	Feb. 23, 2025		
SIO	Feb. 23, 2025	March 9, 2025	June 7, 2023
USNO			May 18, 2024
WHU		Feb. 22, 2025	Feb. 22, 2025
GRG		March 9, 2025	March 7, 2025
CODE	Feb. 23, 2025		Jan. 14, 2025
NGS	Feb. 23, 2025	March 9, 2025	

NIST00USA

Overview AC Usage Equipment Advisories Downloads Photos

Sender	Subject	Date
	NIST station receiver and antenna upgrades	6/29/2022
CODE AC team	Tracking of G04 with Septentrio receiver	2/17/2020
	NIST data delayed since since MJD 58831	12/16/2019
	NIST receiver maintenance after 18.00UTC, MJD58787	10/31/2019
	NIST RINEX data MJD58402/DOY284	10/11/2018
David Maggart	NIST receiver DOY357: bad data ~ first 40 min	3/5/2018

RINEX Observations

RINEX Observations

Zoom 1m 3m 6m YTD 1y All Mar 14, 2025 → Sep 14, 2025

Apr 2025 May 2025 Jun 2025 Jul 2025 Aug 2025 Sep 2025

20k 0

2010 2015 2020

-- GPS -- GLO -- GAL -- BDS -- QZSS

IGS Central Bureau

IGS Data Products

- IGS provides GNSS data recorded at worldwide sites
- IGS produces products to improve GNSS solutions in post-processing
 - GPS orbits/clocks
 - GLONASS orbits
 - IGS station coordinates
 - Earth Rotation Parameters
 - Atmospheric Parameters

Access is via CDDIS – create an account with your email in earthdata

<https://earthdata.nasa.gov/>

https://cddis.nasa.gov/Data_and_Derived_Products/GNSS/GNSS_data_and_product_archive.html



CDDIS – NASA Data Archive

- <https://cddis.nasa.gov/archive/gnss/data/daily/>

- **YYYY/DDD/YYt/mmmmDDD#.YYt.Z**

Example for August 19, 2026

<https://cddis.nasa.gov/archive/gnss/data/daily/2026/231/>

Most files are Unix Compressed – use gunzip to expand
Observation files in the “d” directory are also crx compressed
- use crx2rnx to uncompress

Code	Meaning
YYYY	4-digit year
DDD	3-digit day of year
YY	2-digit year
t	type of data: n = GPS broadcast ephemeris o = observation data in rnx d = observation data in crx
mmm	4-character site monument name
#	sequence number (0, 1, ...) indicating multiple files within a day; typically 0
.Z	UNIX compressed file

Network Status

cddis.nasa.gov/archive/gnss/data/daily/2026/231/26231.V3status														
Personal ASEN5090 ASEN3404 COMPASS Lab AES Links Quantum Projects CONTACT Team Gipsy Is It OK to Commu...														
MGEX Tracking Network Status for 19-Aug-26 260819 26231 GPS Week 2432 Day 4 As of date: Sep 14 2026 13:00:09														
Mon. ID	Full Mon. ID	RNX Ver.	Dly (H)	Dly (M)	V	Receiver Type	Antenna Type	Mkr. Name	Marker Number	Obs. Type	Types G R E C J S I	MD5 Checksum		
ABMF	ABMF00GLP	3.04	56	38	9	SEPT POLARX5	TRM57971.00	NONE	ABMF 97103M001	M	X X X X X X X	1b2c0759b9db46c7711dab1df6cd4329		
ABPO	ABPO00MDG	3.04	0	20	1	SEPT POLARX5	ASH701945G_M	SCIT	ABPO 33302M001	M	X X X X X	35a384814751d583cbf776201308b088		
AC23	AC2300USA	3.04	1	52	1	SEPT POLARX5	TRM59800.99	SCIT	AC23 49296M001	M	X X X X X	5f0bc0133eeddb6d0e804cb7c4a7e022		
AC24	AC2400USA	3.04	1	12	1	SEPT POLARX5	TRM159800.00	SCIT	AC24 49239M001	M	X X X X X X	c23c3090a8f2516771bfc90202820287		
ACRG	ACRG00GHA	3.05	0	18	4	JAVAD TRE_3S	SEPCH0KE_B3E6	NONE	ACRG 32904M001	M	X X X X X X	5802762832e68ad456fa8c6ebd1cd5eb		
ACSO	ACSO00USA	3.04	1	32	1	SEPT POLARX5	TRM59800.80	SCIT	ACSO 49297M001	M	X X X X X	38d34fcadb46e580d4ef6528fe0e6b3f		
ADIS	ADIS00ETH	3.04	201	54	1	LEICA GR50	LEIAR20	LEIM	ADIS 31502M001	M	X X X X X	7386d3f5fb95dfbcdaa579bc58dc9c79		
AGGO	AGGO00ARG													
AIRA	AIRA00JPN	3.02	0	36	1	TRIMBLE ALLOY	TRM59800.00	SCIS	AIRA 21742S001	M	X X X X	c95a2738a420c0bf57705ef6602916ed		
AJAC	AJAC00FRA													
AL2H	AL2H00CAN	3.05	0	06	1	SEPT POLARX5	TRM59800.00	SCIS	AL2H 40129M003	M	X X X X X	0dee042917ba6537339a298a00b5a9f6		
ALB4	ALB400CAN	3.03	0	20	1	JAVAD TRE_G3TH DELTA	TPSCR.G3	NONE	ALB4 40129M008	M	X X X	f7d457f0da5e899b9aed17ab2b960f8		
ALBH	ALBH00CAN	3.03	0	15	1	SEPT POLARX5	TRM59800.00	SCIS	ALBH 40129M003	M	X X X	8c620491066e5a5d5b59fc65b148715e		
ALG2	ALG200CAN	3.05	2	25	1	SEPT POLARX5	NOV750.R4	NOVS	ALG2 40104M005	M	X X X X	5949989fbc25b30aa78c653b7898a773		
ALG3	ALG300CAN													
ALGO	ALGO00CAN	3.05	2	25	1	SEPT POLARX5	AOAD/M_T	NONE	ALGO 40104M002	M	X X X X X	84b5fbfa054a8c2785254363c753f450		
ALIC	ALIC00AUS	3.04	0	24	1	SEPT POLARX5	TWIVC6050	NONE	ALIC 50137M001	M	X X X X X X	5387e37f61c6460a2106a03b77f26073		
AMC4	AMC400USA	3.04	0	25	1	SEPT POLARX5TR	TPSCR.G5C	NONE	AMC4 40472S005	M	X X X X X	0406ac0f5828ab5aeeed66750fc7ea840		
ANK2	ANK200TUR	3.04	1	03	1	TRIMBLE NETR9	TRM115000.00	NONE	ANK2 20818M001	M	X X X X X X	bb2e8490ea19ab6facc9a2e94ae701ee		
ANKR	ANKR00TUR													
ANMG	ANMG00MYS													
ANTF	ANTF00CHL	3.04	281	45	1	STONEX SC2200	STXSA1500	STXG	ANTF 41780M001	M	X X X X X	4b829ce6325d300809b4f73b49887178		
AREG	AREG00PER	3.04	0	28	1	SEPT POLARX5	TRM59800.00	NONE	AREG 42202M008	M	X X X X X	a0f7f10f70ccdb6f9177bcbadd5ce635		
AREQ	AREQ00PER	3.04	0	25	1	SEPT POLARX5	JAVRINGANT_DM	NONE	AREQ 42202M005	M	X X X X X	93442b5f81c4bcdb4f889427ae852139		
ARHT	ARHT00ATA	3.04	0	10	1	JAVAD TRE_3 DELTA	JAVRINGANT_DM	SCIS	ARHT 66001M005	M	X X X X X X X	5869d168463a9f18a4d4f96468097781		
ARUC	ARUC00ARM													
ASCG	ASCG00SHN	3.04	56	43	9	TRIMBLE ALLOY	TRM59800.00	NONE	ASCG 30602M004	M	X X X X X X	6a45287b62cfcc87ea72d0e476313697		
ATLI	ATLI00CAN	3.05	0	11	1	SEPT POLARX5	SEPCH0KE_B3E6	SPKE	ATLI 987009	M	X X X X X X X	a243310a7708246ca8cf101dfb94468a		
ATRU	ATRU00KAZ													
AUCK	AUCK00NZL	3.05	0	11	1	TRIMBLE ALLOY	TRM115000.00	NONE	AUCK 50209M001	M	X X X X X X	b9e7c2912fe767843786dab8415042a2		
AUT0	AUT000USA													
BADG	BADG00RUS	3.04	0	57	1	JAVAD TRE_3 DELTA	JAVRINGANT_DM	JVDM	BADG 12338M002	M	X X X X	6fa8beefa675cee1592adbabb32c8c3e		
BAIE	BAIE00CAN	3.03	0	15	1	TPS NET-G5	TPSCR.G3	NONE	BAIE 40151M001	M	X X	e5e7a04d44eb70d2f58eafb099f03fc		
BAKE	BAKE00CAN	3.03	0	15	1	SEPT POLARX5S	TPSCR.G3	NONE	BAKE 40152M001	M	X X X	3f8298ff856837576e0d991b939b2b70		
BAKO	BAKO00IDN	3.03	201	54	1	LEICA GR50	LEIAR20	LEIM	BAKO 23101M002	M	X X X X X X X X	0754ae5e6bc76abf7ee16df62c950956		
BAMF	BAMF00CAN	3.05	219	11	1	SEPT POLARX5	SEPCH0KE_B3E6	SPKE	BAMF 40190M001	M	X X X X	52086826f6073e5338bb291a61f4cccc		
BAUT	BAUT00DEU	3.05	5	43	4	JAVAD TRE_3S DELTA	LEIAR25.R3	LEIT	BAUT 14102M001	M	X X X X X	ab4d02fb57f75c1285da2d232a5e08d2		
BCOV	BCOV00CAN													
BCRK	BCRK00CAN	3.05	0	11	1	SEPT POLARX5	SEPCH0KE_B3E6	SPKE	BCRK	M	X X X X X X X	dbcc3ac922ce20660107e99f71265c77		
BELE	BELE00BRA	3.05	3	38	1	TRIMBLE NETR8	TRM115000.00	NONE	BELE 41622M001	M	X X	99545e218d757a1769893b11533fd435		
BIK0	BIK000KGZ	3.05	108	12	4	SEPT POLARX5	JAV_RINGANT_G3T	NONE	BIK0 12358M003	M	X X X X X X X X	4ad01232eb635784fee93230993d473b		
BILL	BILL00USA	3.04	3	27	1	SEPT POLARX5	TRM59800.99	SCIT	BILL 49916M001	M	X X X X X X X	b5fe445783cb78ed567935092c50d813		
BOAV	BOAV00BRA	3.05	4	43	1	TRIMBLE NETR9	TRM115000.00	NONE	BOAV 41636M001	M	X X X X X	05195ad531d3cac40fb3528c76fc0047		
BOGI	BOGI00POL	3.05	3	51	1	TPS NET-G5	TPSCR.G5	TPSH	BOGI 12207M003	M	X X X X	6833f4e323d3e037314a1ae7dfe0a31a		
BOGT	BOGT00COL	3.04	0	19	1	JAVAD TRE_3 DELTA	JAVRINGANT_DM	NONE	BOGT 41901M001	M	X X X X	1ab7daca6a1c707ee10838393deab6fe		
BOR1	BOR100POL	3.02	0	28	1	TRIMBLE NETR9	TRM59800.00	NONE	BOR1 12205M002	M	X X X X X X X	9186ef8d7206f856e610dca862c20def		
BPEB	BPEB00CAN	3.05	137	16	1	SEPT POLARX5	SEPCH0KE_B3E6	SPKE	BPEB	M	X X X X	a5dd238b7802c86d7ac2ba4b94516ffe		
BRAZ	BRAZ00BRA	3.05	4	33	1	LEICA GR50	LEIAR10	NONE	BRAZ 41606M001	M	X X X X X	d08c53796da9f648e9e237a8d441bbbb		
BREW	BREW00USA	3.04	0	10	1	SEPT POLARX5TR	ASH701945G_M	SCIT	BREW 40473M001	M	X X X X X X	6428f243e006870b053bfb01e09f1e20		

Receiver Independent Exchange Format (RINEX)

- GPS station converts receiver measurements to RINEX format
- ASCII files
- File types
 - Observation Data file (O)
 - Navigation Message file (Broadcast ephemeris data) (N)
 - Meteorological Data file (M)
- Version 3.0
 - <https://files.igs.org/pub/data/format/rinex305.pdf>

RINEX 3.05

2 THE PHILOSOPHY AND HISTORY OF RINEX

The first proposal for the *Receiver Independent Exchange Format; RINEX* was developed by the Astronomical Institute of the University of Bern for the easy exchange of the Global Positioning System (GPS) data to be collected during the first large European GPS campaign EUREF 89, which involved more than 60 GPS receivers of 4 different manufacturers. The governing aspect during the development was the following fact:

Most geodetic processing software for GPS data use a well-defined set of observables:

- the **carrier-phase measurement** at one or both carriers (actually being a measurement on the beat frequency between the received carrier of the satellite signal and a receiver-generated reference frequency).
- the **pseudorange (code) measurement**, equivalent to the difference of the time of reception (expressed in the time frame of the receiver) and the time of transmission (expressed in the time frame of the satellite) of a distinct satellite signal.
- the **observation time**, being the reading of the receiver clock at the instant of validity of the carrier-phase and/or the code measurements.

Usually the geodetic processing software assumes that the observation time is valid for both the phase **and** the code measurements, **and** for all satellites observed.

RINEX Old file naming convention

- `ssssdddf.yy`
 - `ssss` – 4 digit station identification number
 - `ddd` – day of year
 - `f` – file number
 - `f = 0` – this is a daily file
 - `f = a, b, ..., x` – hourly file (`a = 0h0 – 01h`, `b = 01h to 02h`, etc)
 - `yy` – 2 digit year
 - `t` – file type
 - `O` – observation data file
 - `N` – navigation message file
 - `M` – meteorological file

nist2440.220



RINEX 3.05 Details

4.5 Satellite numbers

For clear unambiguous identification, individual satellites are identified in RINEX files by two-digit number indicating the PRN code or the slot number. This number is preceded by one-character system identifier **s** as shown in Figure 1.

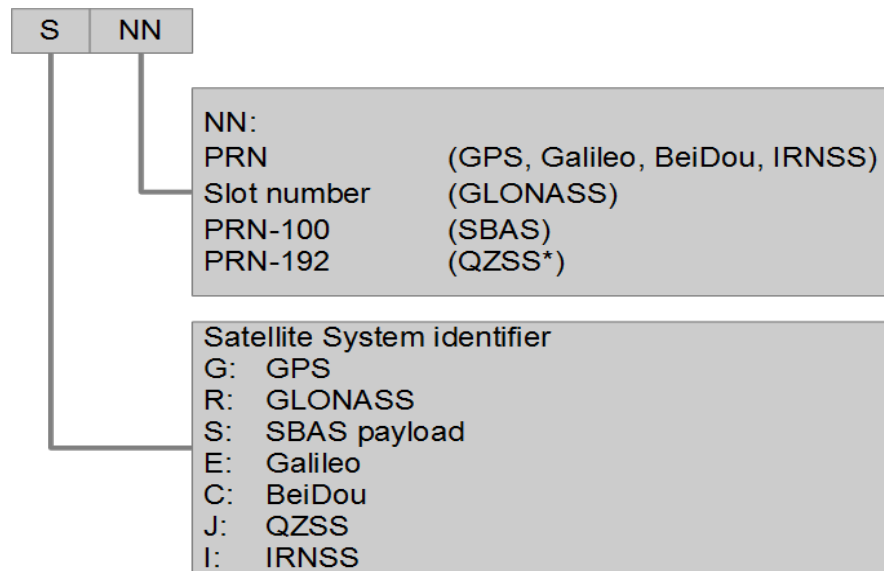


Figure 1: Satellite numbers and Constellation Identifiers

5.1 Long Filenames

From RINEX 3.02 onwards the data filenames are recommended to use the proposed long filenames to be more descriptive, flexible and extensible than the previous short file naming convention.

Table 7 lists example filenames for GNSS observation and navigation files, please note that the source of the data, the start time, the duration, the cadence and the data type are now easily visible. This allows files from the same station over the same time period, different sources, different cadences and with different observation types to be stored together easily.

File Name	Comments
ALGO00CAN_R_20121601000_01H_01S_MO.rnx	Mixed RINEX GNSS observation file containing 1 hour of data, with an observation every second
ALGO00CAN_R_20121601000_15M_01S_GO.rnx	GPS RINEX observation file containing 15 minutes of data, with an observation every second
ALGO00CAN_R_20121601000_01D_30S_MO.rnx	Mixed RINEX GNSS observation file containing 1 day of data, with an observation every 30 seconds
ALGO00CAN_R_20121600000_01D_MN.rnx	RINEX mixed navigation file, containing one day's data

Table 7: Examples of long filenames for RINEX 3 data files

August 19, 2026 NIST observation file
NIST00USA_R_2026**231**0000_01D_**30S**_MO.crx.gz

Observations (Measurements)

5.2.17 Observation codes

Dedicated observation codes are used in RINEX to distinguish individual signals and tracking modes. In order to keep the observation codes short, but still allow for a detailed characterization of the actual signal generation, the observation codes are composed of three characters/digits “**tna**” as detailed in Table 13.

t : observation type	C = pseudo-range	L = carrier phase	D = doppler	S = signal strength	X = channel number
n : band / frequency	1, 2,...,9				
a : attribute	tracking mode or channel, e.g., I , Q , C , P , etc.				

Table 13 : Observation Code Components

Examples:

- **L1C**: C/A code-derived L1 carrier phase (GPS, GLONASS) Carrier phase on E2-L1-E1 derived from C channel (Galileo)
- **C2L**: L2C pseudorange derived from the L channel (GPS)
- **C2X**: L2C pseudorange derived from the mixed (M+L) codes (GPS)

GNSS System	Freq. Band / Frequency	Channel or Code	Observation Codes			
			Pseudo Range	Carrier Phase	Doppler	Signal Strength
GPS	L1/1575.42	C/A	C1C	L1C	D1C	S1C
		L1C (D)	C1S	L1S	D1S	S1S
		L1C (P)	C1L	L1L	D1L	S1L
		L1C (D+P)	C1X	L1X	D1X	S1X
		P (AS off)	C1P	L1P	D1P	S1P
		Z-tracking and similar (AS on)	C1W	L1W	D1W	S1W
		Y	C1Y	L1Y	D1Y	S1Y
		M	C1M	L1M	D1M	S1M
		codeless		L1N	D1N	S1N
	L2/1227.60	C/A	C2C	L2C	D2C	S2C
		L1(C/A) + (P2-P1) (semi-codeless)	C2D	L2D	D2D	S2D
		L2C (M)	C2S	L2S	D2S	S2S
		L2C (L)	C2L	L2L	D2L	S2L
		L2C (M+L)	C2X	L2X	D2X	S2X
		P (AS off)	C2P	L2P	D2P	S2P
		Z-tracking and similar (AS on)	C2W	L2W	D2W	S2W
		Y	C2Y	L2Y	D2Y	S2Y
		M	C2M	L2M	D2M	S2M
		codeless		L2N	D2N	S2N
	L5/1176.45	I	C5I	L5I	D5I	S5I
		Q	C5Q	L5Q	D5Q	S5Q
		I+Q	C5X	L5X	D5X	S5X

Table 14 : RINEX Version 3.05 GPS Observation Codes

RINEX OBS File Header

3.04	OBSERVATION DATA	M	RINEX VERSION / TYPE
ssrcrin-16.3.0x			20260819 000000 UTC PGM / RUN BY / DATE
(00000000581825134591111004)	Septentrio proprietary		COMMENT
NIST			MARKER NAME
0			MARKER NUMBER
GEODETIC			MARKER TYPE
nist	nist		OBSERVER / AGENCY
3069423	SEPT POLARX5TR	5.7.0	REC # / TYPE / VERS
NDE11330004	NOV750.R4	NONE	ANT # / TYPE
-1288399.0601	-4721697.8747	4078626.5526	APPROX POSITION XYZ
0.0000	0.0000	0.0000	ANTENNA: DELTA H/E/N
G	23 X1 C1C L1C D1C S1C C1W S1W C2W L2W D2W S2W C2L L2L		SYS / # / OBS TYPES
	D2L S2L C5Q L5Q D5Q S5Q C1L L1L D1L S1L		SYS / # / OBS TYPES
E	21 X1 C1C L1C D1C S1C C6X L6X D6X S6X C5Q L5Q D5Q S5Q		SYS / # / OBS TYPES
	C7Q L7Q D7Q S7Q C8Q L8Q D8Q S8Q		SYS / # / OBS TYPES
S	9 X1 C1C L1C D1C S1C C5I L5I D5I S5I		SYS / # / OBS TYPES
R	21 X1 C1C L1C D1C S1C C1P L1P D1P S1P C2P L2P D2P S2P		SYS / # / OBS TYPES
	C2C L2C D2C S2C C3Q L3Q D3Q S3Q		SYS / # / OBS TYPES
C	25 X1 C1P L1P D1P S1P C5P L5P D5P S5P C2I L2I D2I S2I		SYS / # / OBS TYPES
	C7I L7I D7I S7I C6I L6I D6I S6I C7D L7D D7D S7D		SYS / # / OBS TYPES
J	25 X1 C1C L1C D1C S1C C2L L2L D2L S2L C5Q L5Q D5Q S5Q		SYS / # / OBS TYPES
	C1L L1L D1L S1L C1Z L1Z D1Z S1Z C5P L5P D5P S5P		SYS / # / OBS TYPES
I	5 X1 C5A L5A D5A S5A		SYS / # / OBS TYPES
SEPTENTRIO RECEIVERS OUTPUT ALIGNED CARRIER PHASES.			COMMENT
NO FURTHER PHASE SHIFT APPLIED IN THE RINEX ENCODER.			COMMENT
G	L1C		SYS / PHASE SHIFT
G	L2W		SYS / PHASE SHIFT
G	L2L	0.00000	SYS / PHASE SHIFT
G	L5Q	0.00000	SYS / PHASE SHIFT
G	L1L	0.00000	SYS / PHASE SHIFT



RINEX Measurements and Format

- C or P = pseudorange (m)
 - L = carrier phase (cycles)
 - D = Doppler (Hz)
 - S = Signal Strength (Units vary)
-
- Format of observations is F14.3 + LLI + Signal Strength Indicator

C1C (m) : 20771165.74249

L1C (cyc) 109153263.03849



RINEX OBS

Body

from Header

G	23	X1	C1C	L1C	D1C	S1C	C1W	S1W	C2W	L2W	D2W	S2W	C2L	L2L	SYS	/	#	/	OBS	TYPES
			D2L	S2L	C5Q	L5Q	D5Q	S5Q	C1L	L1L	D1L	S1L								

> 2026 08 19 00 00 0.0000000 0 36

G03 7.000 22503870.479 7 118258656.23007 -3429.735 7 46.878
 22503870.403 7 43.646 22503874.756 7 92149623.93007 -2672.523 7
 43.646 22503874.981 6 92149627.93206 -2672.550 6 40.742 22503874.447
 7 88310054.99907 -2561.198 7 46.129

G04 8.000 20218540.439 7 106249208.03207 553.284 7 47.692
 20218540.171 6 40.935 20218540.464 6 82791616.84106 431.132 6
 40.935 20218540.778 8 82791623.84108 431.106 8 49.321 20218543.879
 8 79341987.59708 413.126 8 53.166 20218540.093 8 106249207.03208
 553.270 8 48.131

RINEX Composite NAV File

A composite file of all
ephemerides for the day
is constructed
based on data from
multiple stations

(Version 2)

```

2                                NAVIGATION DATA                                RINEX VERSION / TYPE
CCRINEXN V1.6.0 UX  CDDIS                                20-AUG-26 23:31      PGM / RUN BY / DATE
IGS BROADCAST EPHEMERIS FILE                                COMMENT
0.1025D-07  0.2235D-07 -0.5960D-07 -0.1192D-06          ION ALPHA
0.1004D+06  0.1311D+06 -0.6554D+05 -0.3932D+06          ION BETA
0.000000000000D+00 0.888178419700D-15    503808      2432 DELTA-UTC: A0,A1,T,W
18                                                    LEAP SECONDS
                                                    END OF HEADER
1 26  8 19  0  0  0.0 0.193571206182D-03-0.932232069317D-11 0.000000000000D+00
0.114000000000D+03-0.376875000000D+02 0.450054460860D-08-0.218322331505D+01
-0.182911753654D-05 0.188447581604D-02 0.471808016300D-05 0.515359493446D+04
0.259200000000D+06-0.409781932831D-07 0.127484164444D+00 0.819563865662D-07
0.956986158269D+00 0.286906250000D+03 0.172485168883D+00-0.820855620503D-08
-0.177507393895D-09 0.100000000000D+01 0.243200000000D+04 0.000000000000D+00
0.200000000000D+01 0.000000000000D+00-0.884756445885D-08 0.114000000000D+03
0.252018000000D+06 0.400000000000D+01 0.000000000000D+00 0.000000000000D+00
2 26  8 19  0  0  0.0 0.159730669111D-03 0.704858393874D-11 0.000000000000D+00
0.157000000000D+03-0.248750000000D+02 0.448732977244D-08-0.844302047838D+00
-0.124424695969D-05 0.169594280887D-01 0.465288758278D-05 0.515371703148D+04
0.259200000000D+06-0.301748514175D-06-0.342231880932D-01-0.134110450745D-06
0.961815590246D+00 0.293843750000D+03-0.722472067622D+00-0.837713465557D-08
-0.578595529398D-10 0.100000000000D+01 0.243200000000D+04 0.000000000000D+00
0.200000000000D+01 0.000000000000D+00-0.176951289177D-07 0.157000000000D+03
0.252018000000D+06 0.400000000000D+01 0.000000000000D+00 0.000000000000D+00
3 26  8 19  0  0  0.0 0.315596349537D-03-0.200088834390D-10 0.000000000000D+00
0.470000000000D+02-0.518437500000D+02 0.343800034937D-08 0.192775957627D+01
-0.273436307907D-05 0.696313160006D-02 0.865571200848D-05 0.515375367165D+04
0.259200000000D+06 0.204890966415D-07 0.112048278835D+01-0.145286321640D-06
0.995156738215D+00 0.229187500000D+03 0.129150910753D+01-0.755245744731D-08
0.217866217860D-09 0.100000000000D+01 0.243200000000D+04 0.000000000000D+00
0.280000000000D+01 0.000000000000D+00 0.139698386192D-08 0.470000000000D+02
0.252018000000D+06 0.400000000000D+01 0.000000000000D+00 0.000000000000D+00

```

RINEX Composite NAV File

- Note - sometimes there are updates not on the 2hr boundaries
- Corrected uploads are tagged with times 16 s earlier than the time of the incorrect upload. Multiple uploads move forward by additional 16s shifts
- “read_clean_GPSbroadcast.m” takes care of removing the bad uploads.

```
3 26 8 19 1 59 44.0 0.315454322845D-03-0.200088834390D-10 0.000000000000D+00
0.900000000000D+01-0.434687500000D+02 0.338371237377D-08 0.297550750407D+01
-0.214949250221D-05 0.696215755306D-02 0.873394310474D-05 0.515375542641D+04
0.266384000000D+06 0.316649675369D-07 0.112042954984D+01 0.745058059692D-08
0.995158591733D+00 0.226156250000D+03 0.129155124542D+01-0.742459497847D-08
0.157863718515D-09 0.100000000000D+01 0.243200000000D+04 0.000000000000D+00
0.200000000000D+01 0.000000000000D+00 0.139698386192D-08 0.900000000000D+01
0.261528000000D+06 0.400000000000D+01 0.000000000000D+00 0.000000000000D+00
5 26 8 19 1 59 44.0-0.236096791923D-03-0.341060513165D-12 0.000000000000D+00
0.430000000000D+02-0.438125000000D+02 0.381730186309D-08 0.233791903583D+00
-0.215135514736D-05 0.546725816093D-02 0.785663723946D-05 0.515355220985D+04
0.266384000000D+06-0.152736902237D-06 0.105845125821D+01-0.670552253723D-07
0.981551755148D+00 0.242437500000D+03 0.149416787759D+01-0.777103798064D-08
0.917895376884D-10 0.100000000000D+01 0.243200000000D+04 0.000000000000D+00
0.200000000000D+01 0.000000000000D+00-0.102445483208D-07 0.430000000000D+02
0.264366000000D+06 0.400000000000D+01 0.000000000000D+00 0.000000000000D+00
32 26 8 19 1 59 44.0 0.224197283387D-03 0.193267624127D-10 0.000000000000D+00
0.120000000000D+02-0.141875000000D+02 0.443804200512D-08 0.640829419883D+00
-0.881031155586D-06 0.950230984017D-02 0.664032995701D-05 0.515358444595D+04
0.266384000000D+06-0.171363353729D-06 0.214064671112D+01 0.111758708954D-07
0.969702621949D+00 0.255937500000D+03-0.197480665597D+01-0.814355349741D-08
0.263939565571D-09 0.100000000000D+01 0.243200000000D+04 0.000000000000D+00
0.200000000000D+01 0.000000000000D+00 0.931322574616D-09 0.120000000000D+02
0.265308000000D+06 0.400000000000D+01 0.000000000000D+00 0.000000000000D+00
1 26 8 19 2 0 0.0 0.193504150957D-03-0.932232069317D-11 0.000000000000D+00
0.115000000000D+03-0.379687500000D+02 0.451447376024D-08-0.113251076914D+01
-0.189244747162D-05 0.188433995936D-02 0.519864261150D-05 0.515359666443D+04
0.266400000000D+06-0.186264514923D-07 0.127425641870D+00-0.540167093277D-07
0.956984432026D+00 0.280187500000D+03 0.172000627008D+00-0.821712799065D-08
-0.290726395636D-09 0.100000000000D+01 0.243200000000D+04 0.000000000000D+00
0.200000000000D+01 0.000000000000D+00-0.884756445885D-08 0.115000000000D+03
0.259218000000D+06 0.400000000000D+01 0.000000000000D+00 0.000000000000D+00
```

Precise Orbit and Clock Products

IGS combined solutions

The IGS Analysis Centers (ACs) submit GNSS satellite orbit and satellite and station clock solutions, as well as Earth rotation parameters (ERPs), to the global data centers on sub-daily, daily or weekly schedules. These solutions are combined to form the official IGS products. The IGS orbit, clock and ERP products include:

- IGS ultra-rapid orbits and ERPs: useful for real-time and near-real-time applications, provided four times per day (every six hours). They consist of 24 hours of observed and 24 hours of predicted combined satellite orbits (and ERPs).
- IGS rapid orbits, clocks and ERPs: daily combined solutions available approximately 17 hours after the end of the previous UTC day.
- IGS final orbits, clocks and ERPs: considered the highest quality and the most consistent IGS products, include daily combined satellite orbits. The IGS final ERPs are derived from the terrestrial frame combinations (see https://igs.org/products#terrestrial_frame).

Combined solutions are also provided as part of each IGS reprocessing campaign.

FIN - Final orbits, 2 weeks late
RAP – Rapid orbits
ULT – Ultra rapid orbits

https://igs.org/products/#orbits_clocks

for Aug 19 <https://cddis.nasa.gov/archive/gnss/products/2432>

Organized by GPS week number

IGS Analysis Centers (ACs)

Institution	Abbreviation	Country/Region
Natural Resources Canada	EMR	Canada
Wuhan University	WHU	China
Geodetic Observatory Pecny	GOP-RIGTC	Czech Republic
Space geodesy team of the CNES	GRG	France
European Space Agency/ESOC	ESA/ESOC	Germany
GFZ Helmholtz Centre for Geosciences	GFZ	Germany
Geospacial Information Authority of Japan and Japan Aerospace Exploration Agency	JGX	Japan
Center for Orbit Determination in Europe	CODE	Switzerland
Jet Propulsion Laboratory	JPL	USA
Massachusetts Institute of Technology	MIT	USA
NOAA/National Geodetic Survey	NGS	USA
Scripps Institution of Oceanography	SIO	USA
U.S. Naval Observatory	USNO	USA

Last Updated on 12 Mar 2025 02:07 UTC

References

- Gary Johnston, Anna Riddell, and Grant Hausler, Chapt 33
The International GNSS Service, in Peter J.G. Teunissen and
Oliver Montenbruck (eds.) Springer Handbook of Global Navigation Satellite
Systems, DOI:10.1007/978-3-319-42928-1_33
- Dow, J.M., R.E. Neilan, C. Rizos (2009) The International GNSS Service in a
changing landscape of Global Navigation Satellite Systems. J. Geod. 83 (3–4),
191–198. <http://dx.doi.org/10.1007/s00190-008-0300-3>.

Acknowledgements

ASEN5090 lecture materials have been developed at the University of Colorado Boulder by Professors Penina Axelrad, Dennis Akos, Kristine M. Larson, Jade Morton, and Yang Wang, with additional contributions by many outstanding TA's and students.

Next lecture: ionospheric effects and models